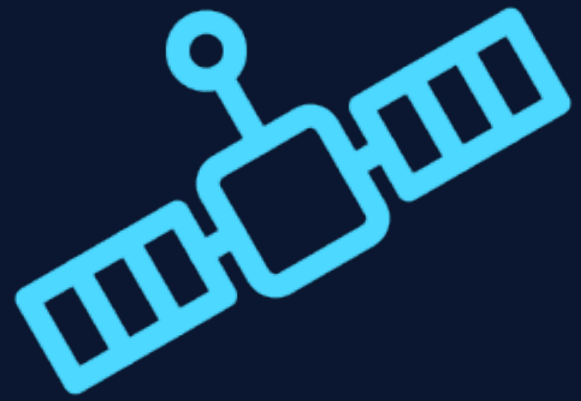
Software Systems · Project concept

# Satellite Predictive Maintenance



Estimating reaction-wheel remaining useful life from sensor history.

1 **Measure**
Collect sensor readings

2 **Estimate**
Track degradation and remaining life

3 **Prioritize**
Support earlier engineering decisions

# Predictive maintenance works across domains.

Use machine measurements and operational context to anticipate degradation instead of reacting after a breakdown.



## Industrial machinery

Motors, gearboxes, factory bearings



## Wind turbines

Rotating shafts, generator bearings



## Satellite systems

Reaction-wheel mechanisms

**The general idea**

**Sensor data** → **Software** → **Maintenance insight**

# NASA Kepler: an operational warning.

A documented case where reaction-wheel deterioration changed a space mission.

**Jul 2012**

## First wheel failed

Kepler lost reaction wheel 2.

**Early 2013**

## Friction symptoms

Increasing friction appeared in wheel 4; engineers attempted life-extension measures.

**May 2013**

## Wheel 4 failed

Only two working wheels remained.



# 3 of 4

wheels required for original precision pointing

**Why forecast?** More lead time to review wheel health, adapt operations and prepare contingencies.

NASA (2013) and NESC/Dennehy (2014). Warnings do not guarantee prevention of failure.

# From Sensor Measurements to Remaining Useful Life

PRONOSTIA bearing measurements enter our software; the output is an estimate of remaining operating time.




Nectoux et al. (IEEE PHM 2012), “PRONOSTIA: An Experimental Platform...” • Bearing benchmark, not satellite flight data.

# AI agents support our development.

Focused coding assistance throughout implementation, testing and integration.



1

## Data & features

Implement dataset readers, validation and sensor-feature extraction.



2

## RUL component

Translate our chosen estimation approach into modular, testable code.



3

## Tests & integration

Generate tests, investigate bugs, refactor and integrate team changes.

**Our team defines the requirements, reviews the code and owns the final result.**

# Selected technical and mission sources.

Sources underpinning the use case and the prototype dataset.

| | | |
|---|---|---|
| 1 | **NASA / Kepler mission** | Two reaction-wheel failures; the original pointing requirement; operational outcome. |
| 2 | **C. J. Dennehy / NASA NESC (2014)** | Increased friction and mitigation efforts on Kepler wheel 4. |
| 3 | **Nectoux et al. / IEEE PHM (2012)** | PRONOSTIA: experimental platform for bearing degradation and RUL research. |
| 4 | **NASA Prognostics Repository** | Distribution point for the FEMTO / PRONOSTIA bearing dataset. |
| 5 | **Muthuganapathy et al. / ASR (2026)** | Reaction-wheel diagnostics using speed, current, temperature and torque telemetry. |